AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 11 · STUDENT LESSON

Visual Composition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Count the objects. Trace the relations. Check the endpoints.

Evaluate visual composition through explicit counts, labels, position and meaningful flow.

Keep one habit: Complete brief → criterion table → reconstructed layouts → three prompt repairs → independent containment.

MAIN LESSON · SECTION 1

Are the right objects in the right relationship?

> A composition can have correct counts while communicating the wrong meaning.

Your current composition goals

I can check object counts, spatial relations and arrow meaning separately.

I can turn an ambiguous visual request into observable composition criteria.

I can distinguish a supplied layout description from an inspected original image.

Recall visible checks and origin

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Image Generation Models: a correct-looking object count does not establish that labels, spatial relations or provenance claims are correct. Give one check for each.

Your end goal

By the end, I can check object counts and spatial relations as separate visual requirements.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

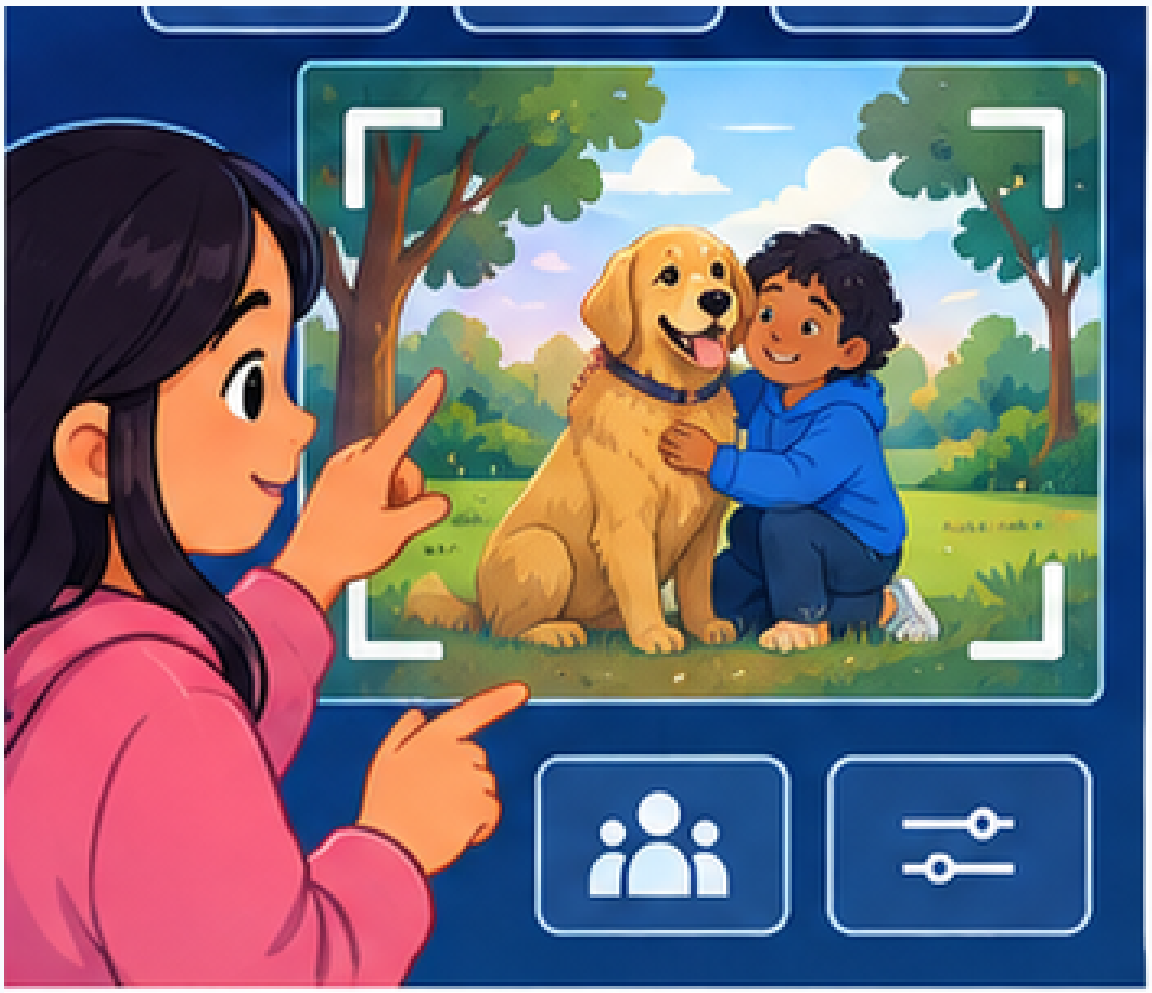
Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 11: Visual Composition

Subject, **setting**, **framing**, **relationship**s, and constraints shape an image request.
</details>



CREATOR MISSION

Diagnose and repair three ambiguous visual prompts.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Define the relationship before judging the picture

> The main brief supplies exact counts, labels and flow endpoints.

What composition adds to object presence

Composition describes how objects are arranged and related: count, relative position, size, containment, visible labels and the source/destination of arrows. “Draw bottles and a beaker” names subjects but leaves key relationships unspecified. A clear brief turns the intended meaning into checks that a partner can apply. A more precise request is still not proof that an output follows it.

The complete main contract

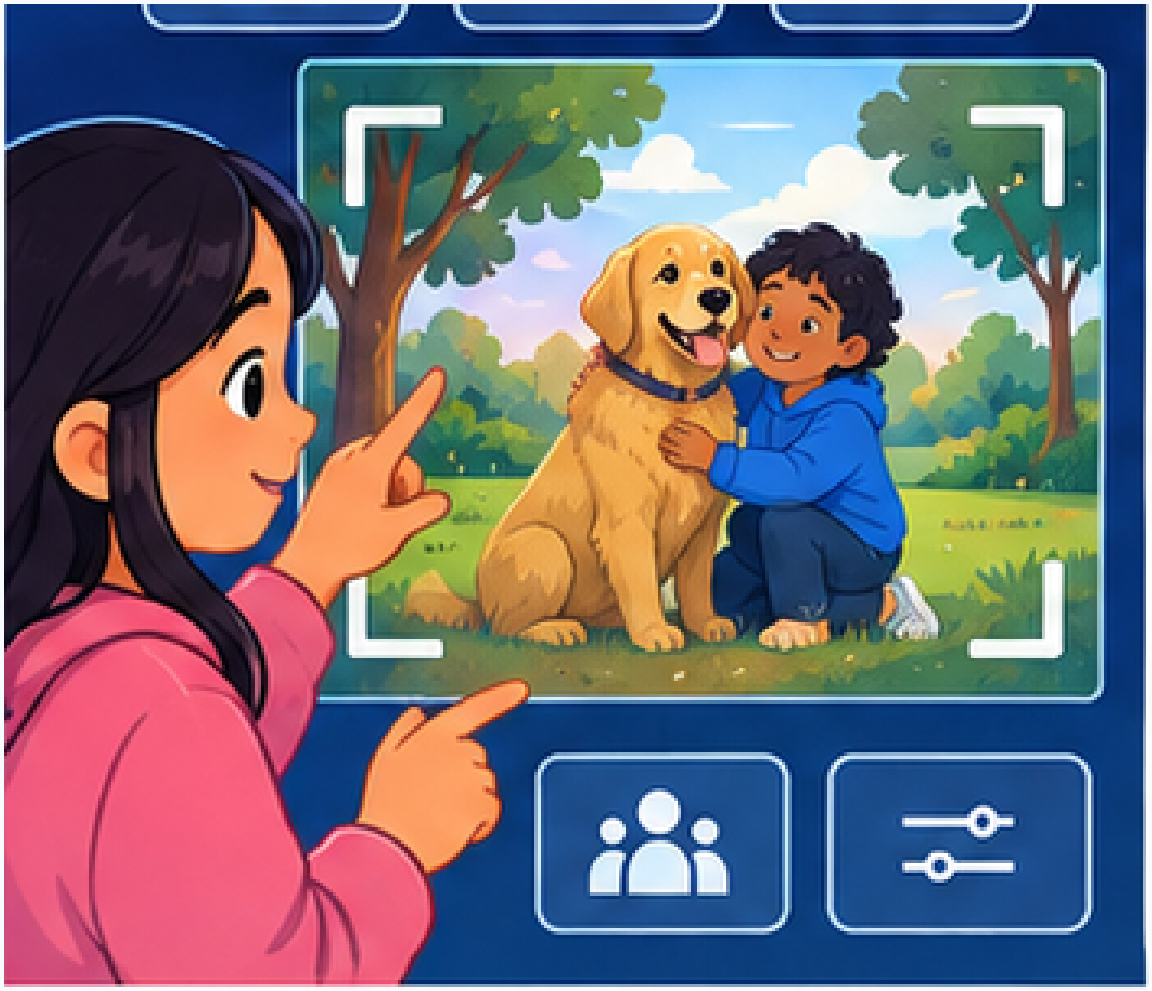
Main fictional layout contract: exactly two labelled bottles B1,B2 left of one beaker; one arrow from the bottle pair toward the beaker. Use the supplied description, not an assumed experiment. Original description places the bottles right of the beaker and draws the arrow from beaker toward bottles. Count, left-of and semantic-flow checks are separate; no chemistry procedure is validated by this layout.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use visual composition to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Check counts, labels, position and flow separately

> Every criterion needs a concrete referent in the description or image.

The complete composition checklist

Check	Evidence required by the brief
Count	Exactly two bottles and one beaker
Labels	B1 and B2 identify the bottles legibly
Position	Both bottles are left of the beaker
Flow	Exactly one arrow starts at the bottle pair and points toward the beaker

Check	Evidence required by the brief
Scope	A layout reconstruction is not an original generator output or chemistry observation

Explain the deciding relation

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Review · Earlier generation and provenance guidance

Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Diagnose and repair three ambiguous visual prompts.
↓
SYSTEM
Subject, setting, framing, relationships, and constraints shape an image request.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Image generation and provenance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.	Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use.	It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

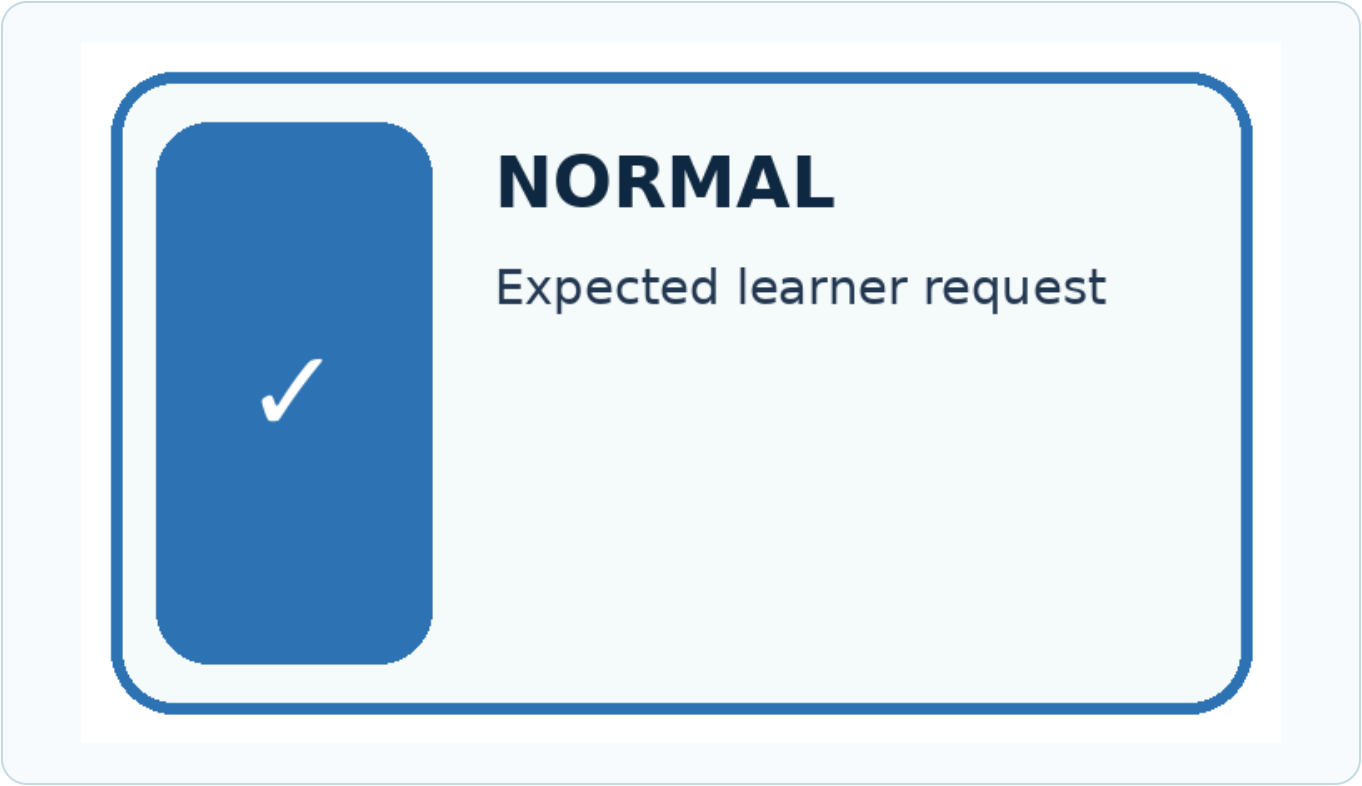
Connect the picture and the explanation

MAIN LESSON · SECTION 4

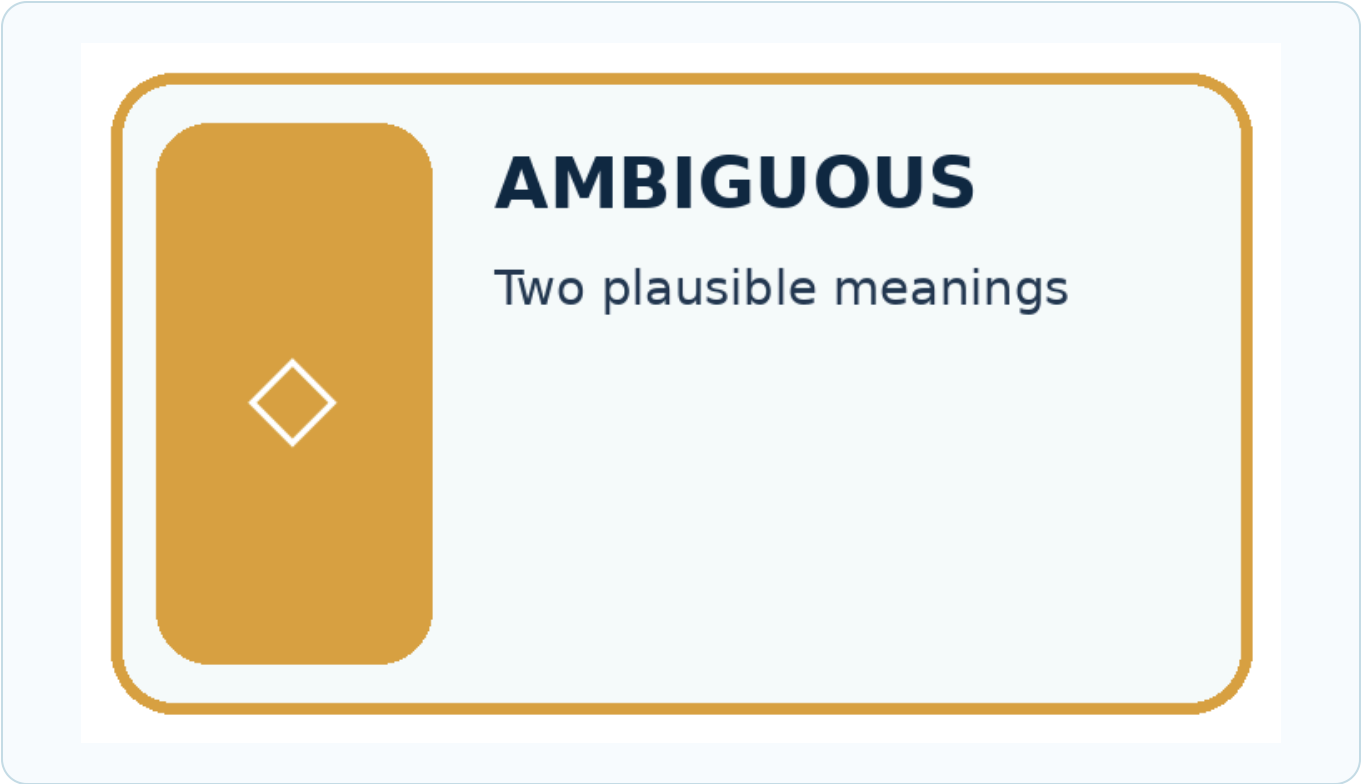
Compare correct subjects with correct relations

> Count, position and meaningful arrow direction can fail independently.

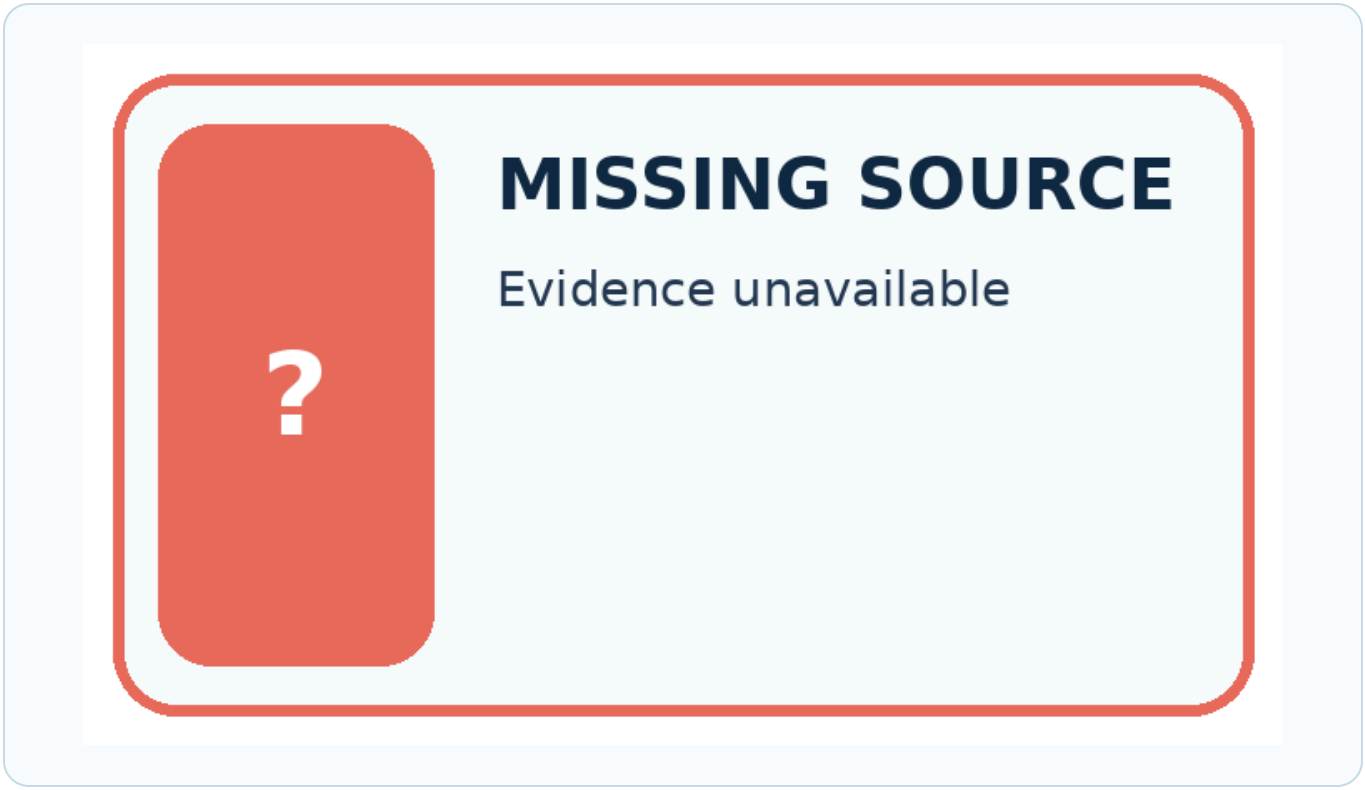
Four pictures with a purpose



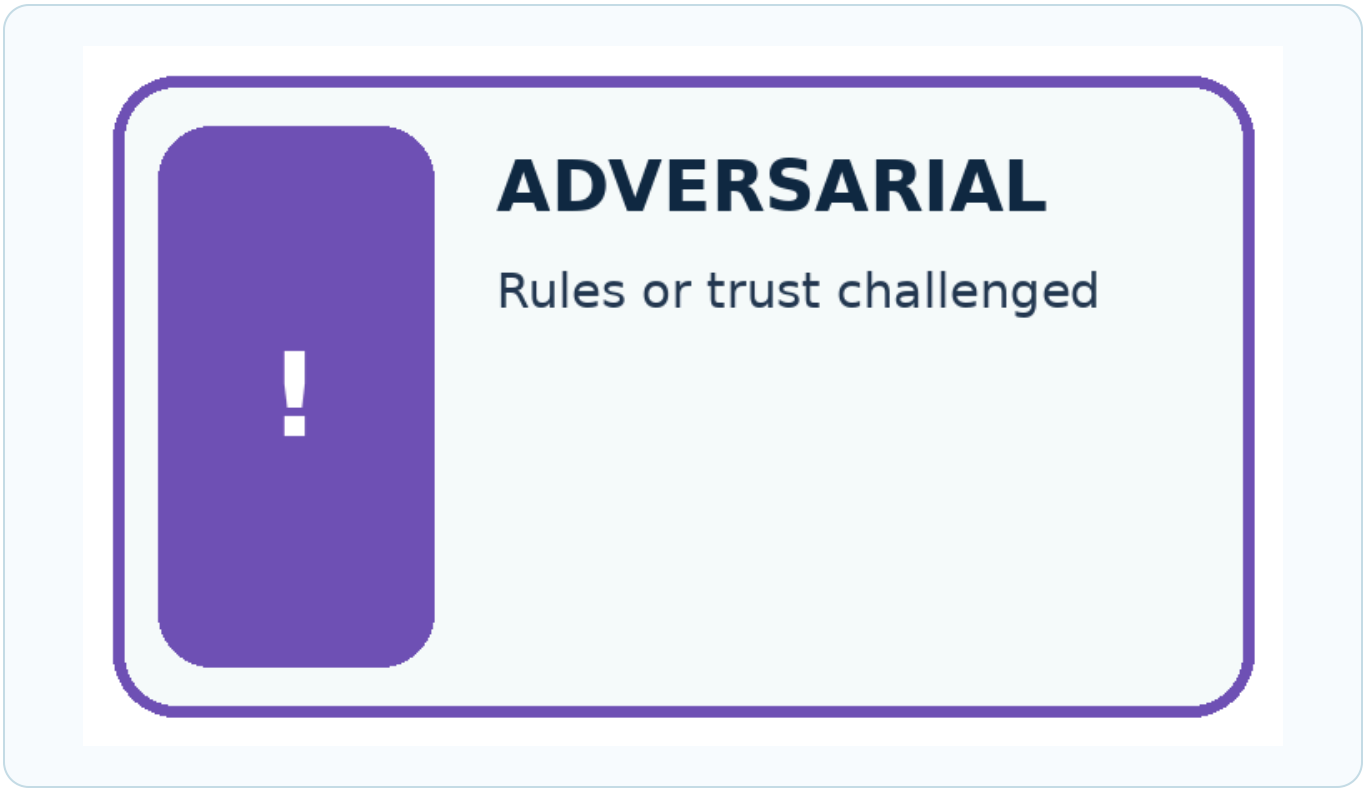
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

▼ Check after predicting

Prediction: Position now passes; arrow direction still fails.

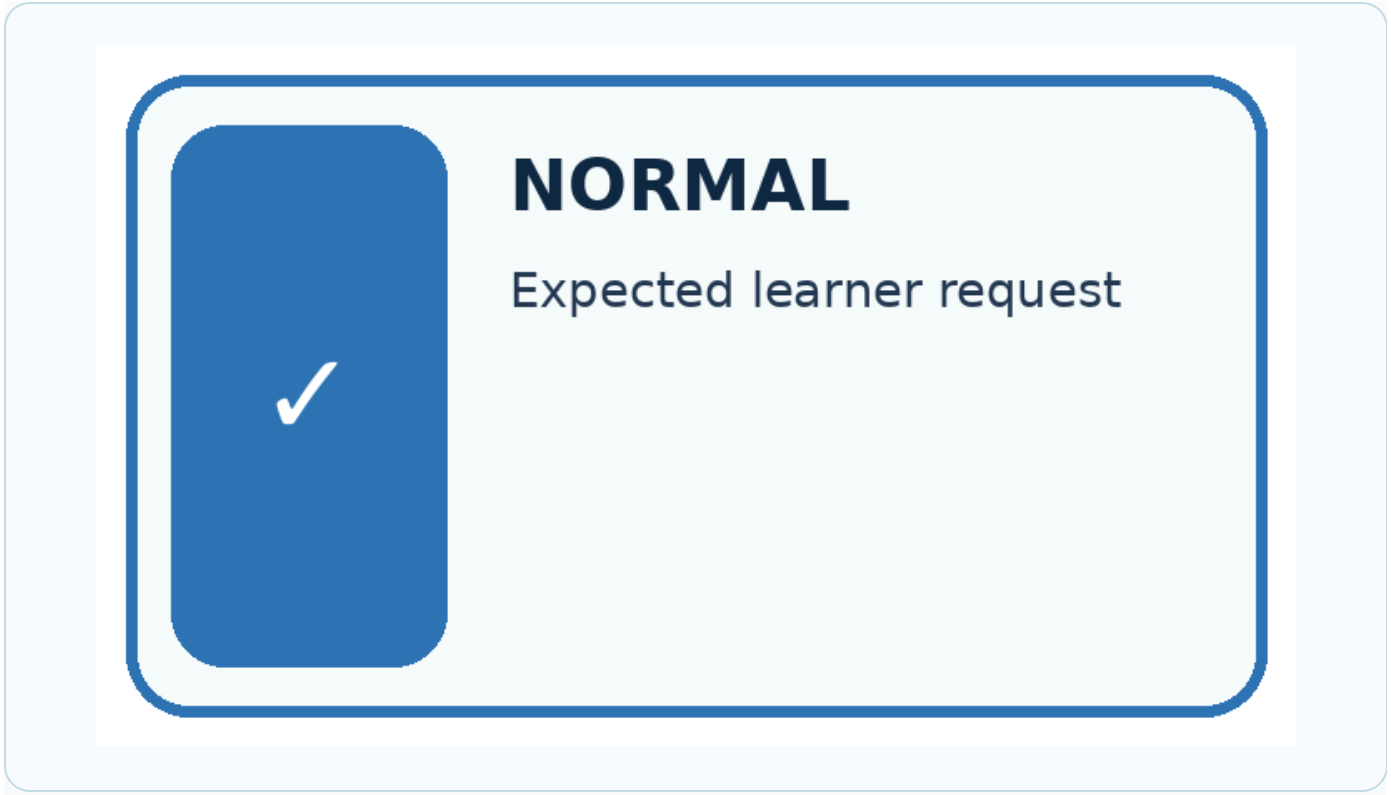
Reason: Changing object placement does not automatically reverse an arrow's semantic source and destination.

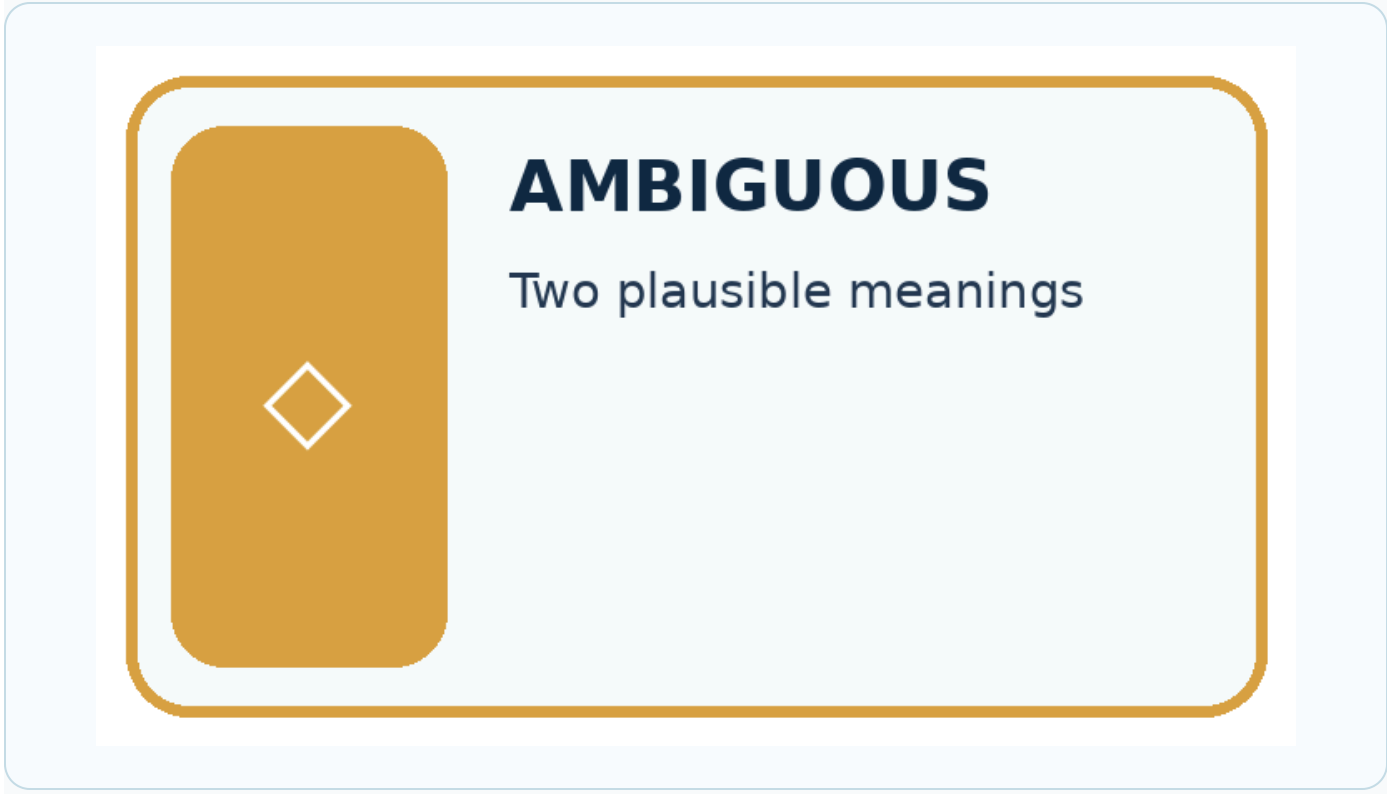
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace one changed relation without assuming another is fixed

> Use the reconstructed layouts alongside the supplied written case.

Read the complete brief and original description

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

Inspect the three disclosed classroom reconstructions

CONSTRUCTED LAYOUTS FROM THE WRITTEN CASE

Required: two bottles left of one beaker; flow from bottles to beaker.

1 · Described original



Counts pass · position fails · semantic flow fails

2 · Move bottles only; keep beaker → bottles



Counts pass · position passes · semantic flow fails

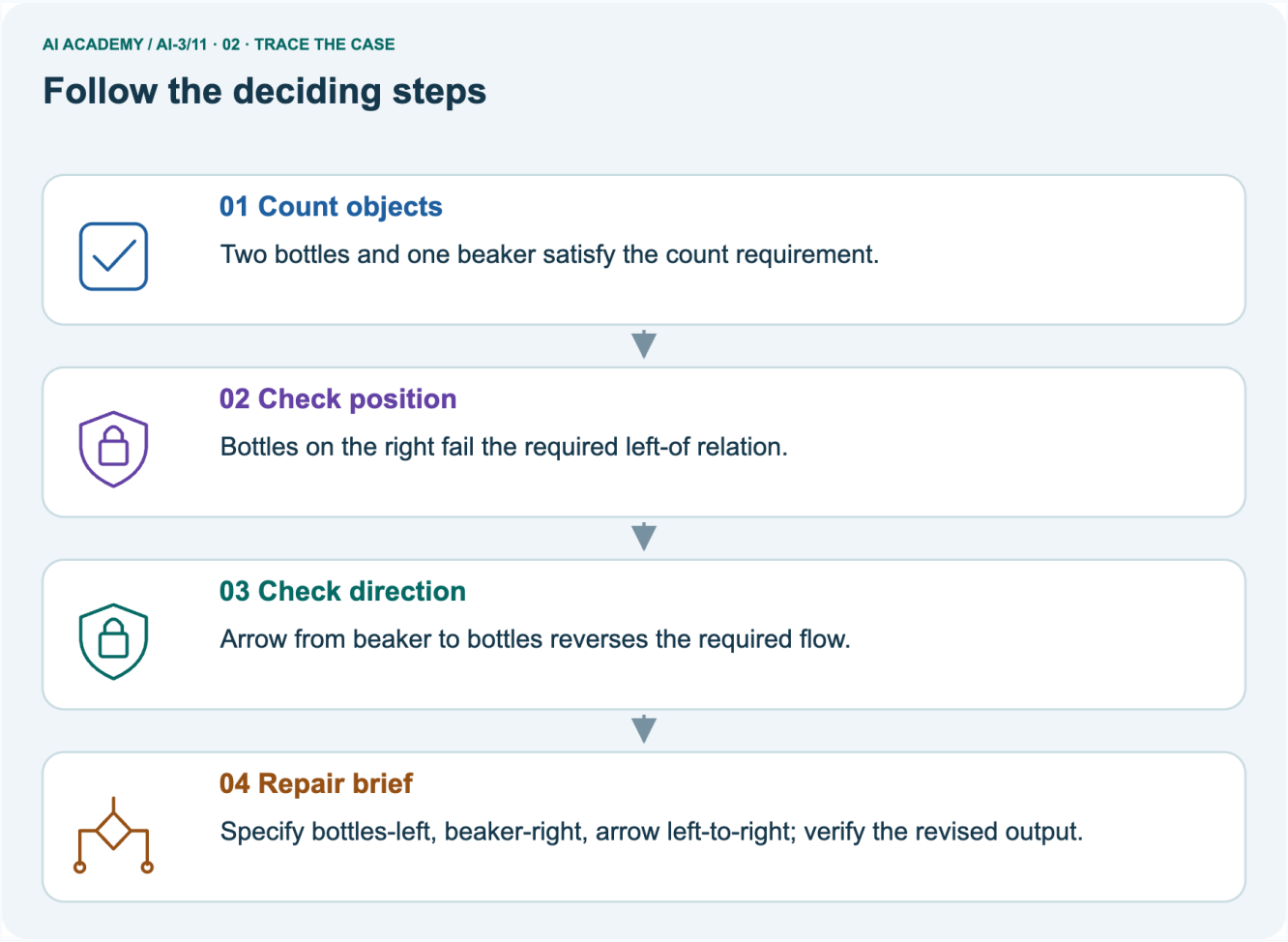
3 · Complete proposed layout repair



Counts pass · position passes · semantic flow passes

These classroom diagrams illustrate descriptions; they are not generator outputs or verified chemistry.

Trace the count, position and flow checks



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count objects	Two bottles and one beaker satisfy the count requirement.
2. Check position	Bottles on the right fail the required left-of relation.
3. Check direction	Arrow from beaker to bottles reverses the required flow.
4. Repair brief	Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Understand named endpoints and image scope

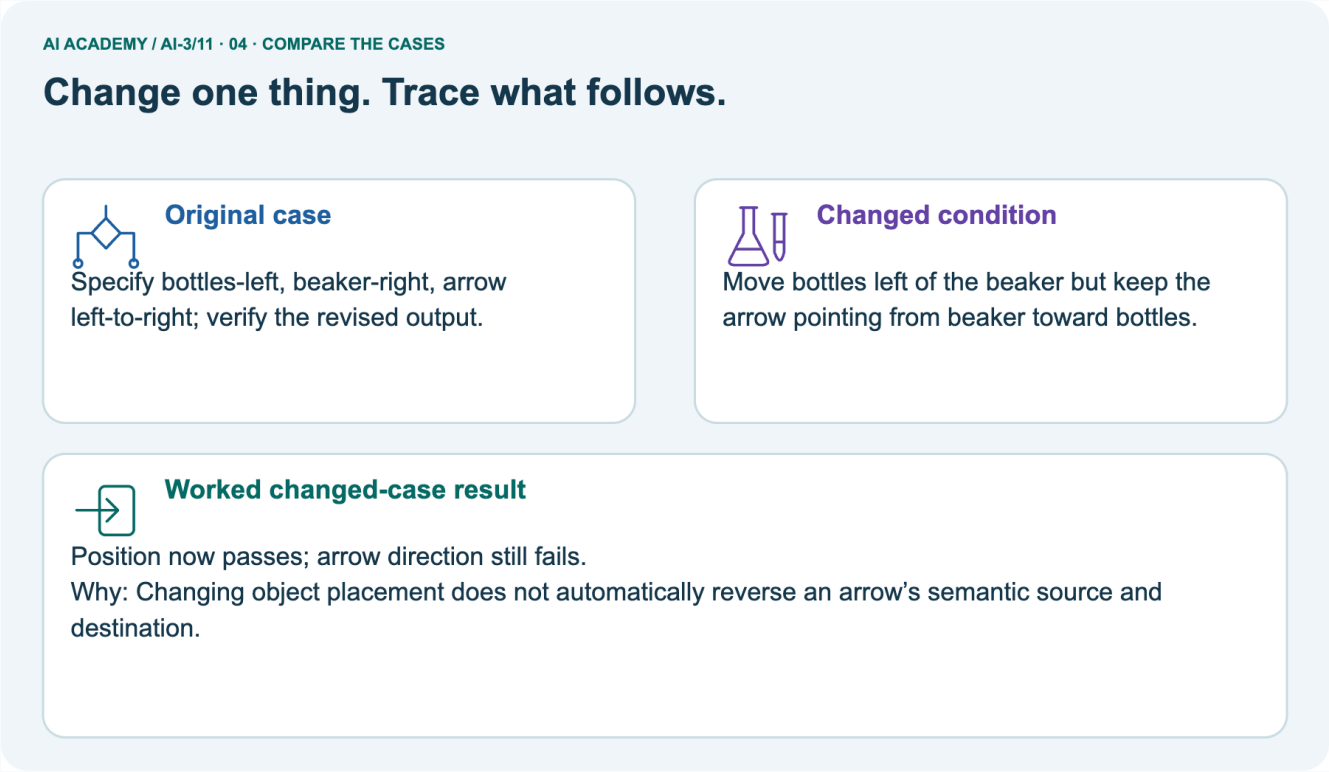
The original arrow can point left-to-right on the page and still be wrong: it starts at the beaker, while the brief requires flow from bottles. After moving bottles left but preserving the beaker→bottles meaning, the arrow points right-to-left. Judge the named endpoints, not direction on the page alone.

The three native diagrams reconstruct the supplied descriptions; the original generated image is not supplied and no model is run.

Predict moving only the bottles

- Change one thing: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain remaining content limits

- A compliant layout does not by itself verify the chemistry or procedure represented; content needs source checking.

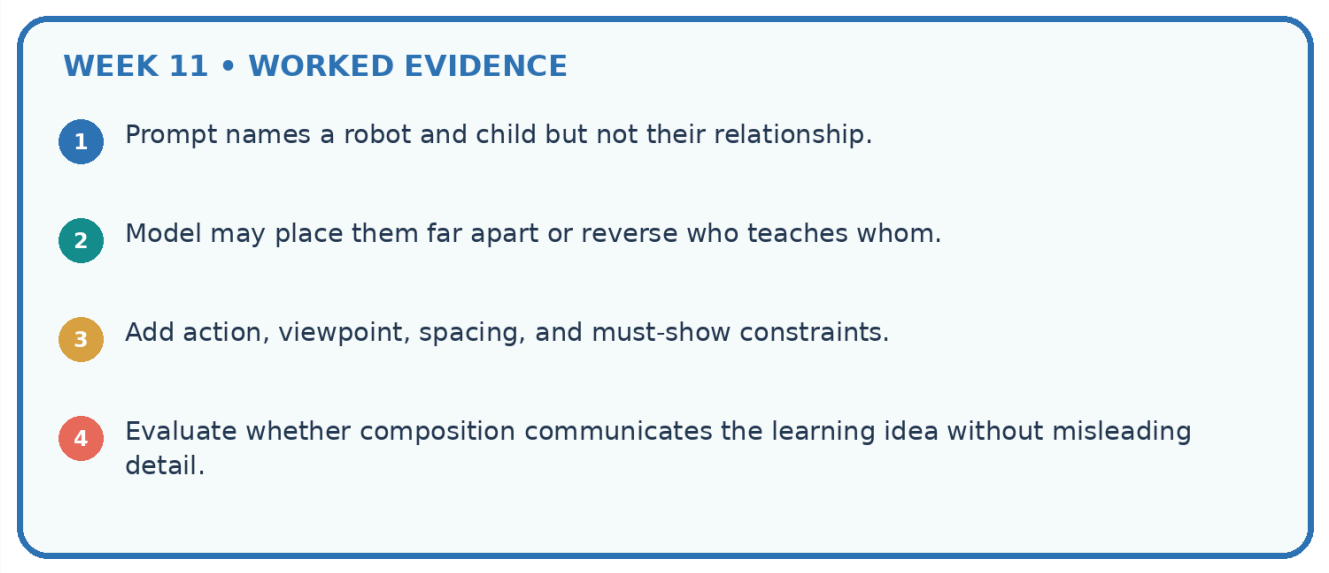
- Read the labelled picture: Which checks pass after that single change, and what still needs repair?

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Prompt names a robot and child but not their **relationship**.

2. Model may place them far apart or reverse who teaches whom.

3. Add action, viewpoint, spacing, and must-show constraints.

4. Evaluate whether composition communicates the learning idea without misleading detail.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

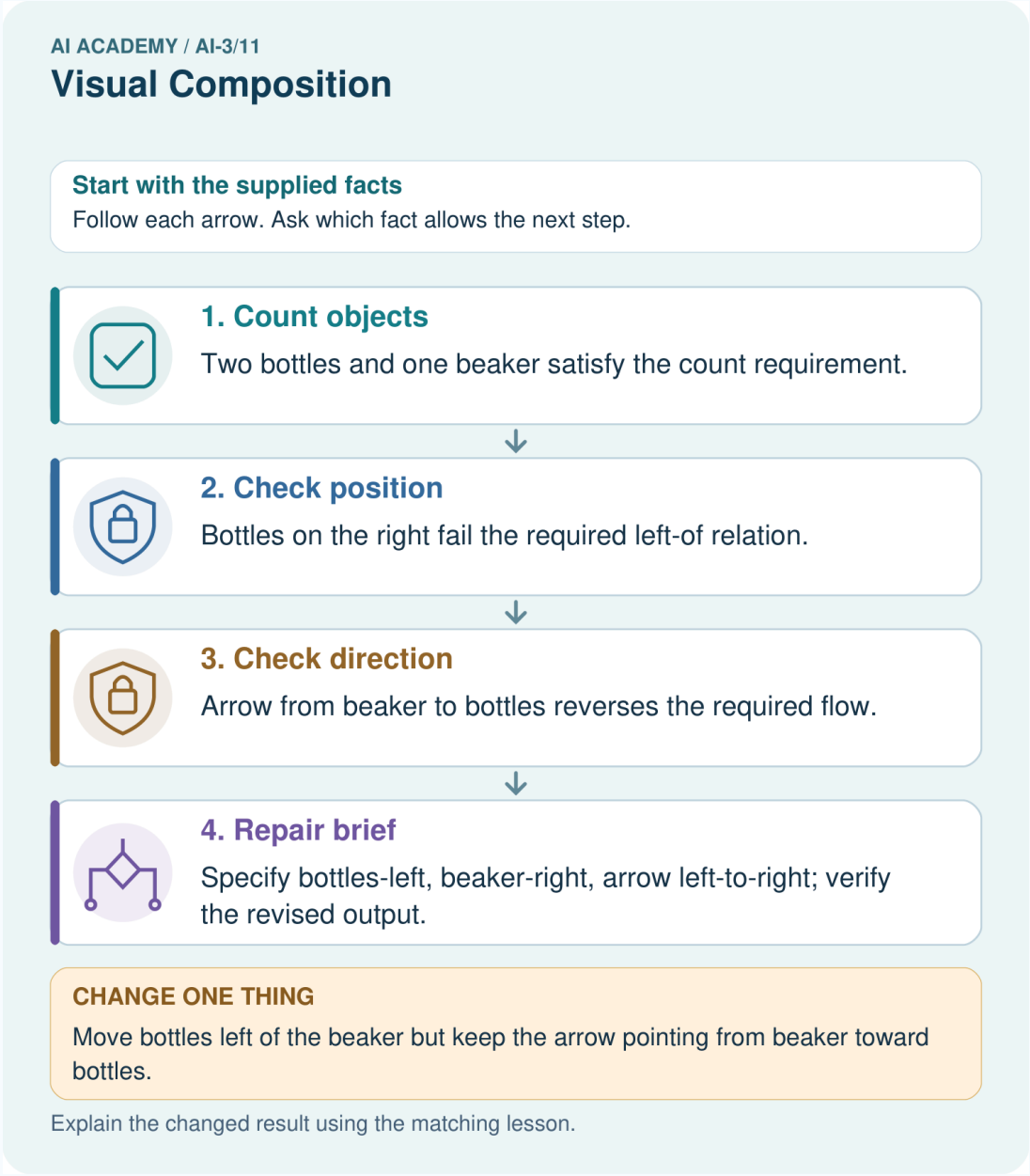
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Visual Composition

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the relation hidden by correct object counts

> Use one reason per criterion and preserve the failed ones.

Repair the count-only argument

Claim to repair: There are two bottles and one beaker, so the layout passes. The count criterion passes in both the original and changed descriptions. Originally position and semantic flow fail. Moving bottles fixes only position; beaker→bottles still reverses the intended flow. A complete proposed repair must satisfy both relations and then be checked against an actual revised output if one is produced.

Design a revealing one-change check

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: More visual adjectives always create a clearer composition. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the layout trace with less help

> Name the object roles rather than checking only an arrow’s page direction.

Current supported task and response

- For the main original and move-bottles-only descriptions, judge count, bottle labels, left-of position and semantic arrow flow separately. Give the passing evidence and the remaining repair. State whether you inspected an original model image or a classroom reconstruction.

My current spatial-relationship reasoning

Explain the pictured one-change result

- Which checks pass after that single change, and what still needs repair?

Explain the picture

My explanation using a pictured step and the given facts

Keep independent work separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Review · Earlier poster/provenance practice

Provenance tracks production history; it does not prove the content is accurate or establish every legal permission.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Image generation and **provenance**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Check your reasoning

Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

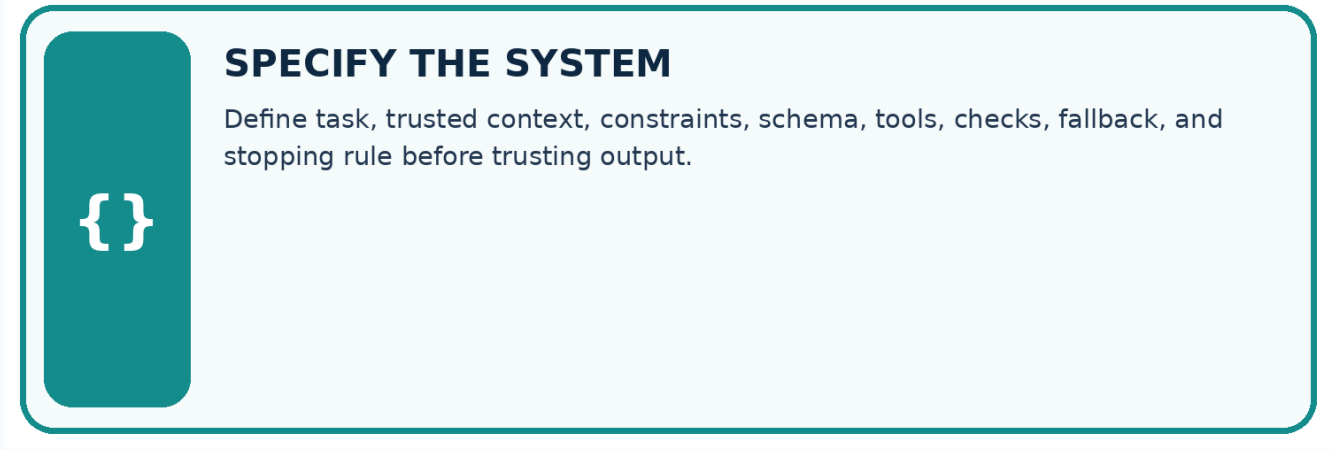
Answer on your own before discussion. If you are unsure, say which step you need help with.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Repair three complete ambiguous briefs

The missing composition element is stated for each card.

The three supplied prompt cards

Prompt card	Complete supplied brief	Missing component to repair
A	Exactly two bottles labelled B1,B2 and one beaker;one arrow bottles→beaker.	Position of the bottle pair relative to beaker
B	Exactly two bottles labelled B1,B2 left of one beaker;show one arrow.	Arrow's semantic source and destination
C	Some labelled bottles left of one beaker;one arrow bottles→beaker.	Exact bottle count

Change one component, sketch and peer-check

1. Three-prompt composition investigation: repair only the named missing component in each complete card. A adds bottles-left/beaker-right; B adds bottles→beaker flow; C replaces some with exactly two. Sketch the expected layout and write separate count, label, position and flow checks before any generation. Exchange sketches with a partner who identifies the named endpoints and remaining ambiguity.

   Compare the original and move-bottles-only reconstructed diagrams using the same checklist. These are prompt plans and paper drawings, not newly generated outputs or evidence that an image model improved. If a later image is actually produced, inspect its criteria and known production record; do not mark an unrun proposal as passed. Keep the stars/containment independent description withheld.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____

> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → RUN → VERIFY → EXPLAIN

Repair three prompts by changing one missing composition element; sketch expected framing before generation and compare.

Use the supplied A/B/C prompt cards. Repair position, semantic arrow endpoints and exact count respectively, preserving other stated conditions. Sketch expected results and peer-check the composition criteria. No actual image generation or improved model result is supplied.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: evaluate the fresh stars-and-containment record

> Object presence and full containment are different checks.

Independent description and scope

A supplied drawing description says three stars are above a triangle, while a circle crosses the edge of a rectangle. The brief requires three stars above the triangle and the circle fully inside the rectangle.

Evaluate this description.

Evaluate only the supplied stars/triangle/circle/rectangle description. No original image or measurements are supplied. State which count and relationship checks the description supports and which containment requirement it contradicts; do not invent hidden geometry

or inspect the classroom bottle reconstruction as if it were this task's image.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A poster puts pale yellow labels on white and hides the main object behind decorations. Suggest two revisions and how to check them.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create age-fit explanatory visuals with labeled purpose and no invented documentary **evidence**.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Give each project diagram a meaning-based checklist

> Keep composition success separate from factual and origin claims.

This week’s concrete contribution

For Verified Study Buddy diagrams, state the teaching purpose and turn its meaning into observable counts, readable labels, containment/order and named arrow source/destination. Distinguish a diagram reconstructed from a description from an original generated output. Verify factual subject matter with approved sources even if the layout passes; a correct arrangement alone cannot establish an experimental result.

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

Keep the checkpoint and human choice

THIS WEEK’S CHECKPOINT Create age-fit explanatory visuals with labeled purpose and no invented documentary **evidence**.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

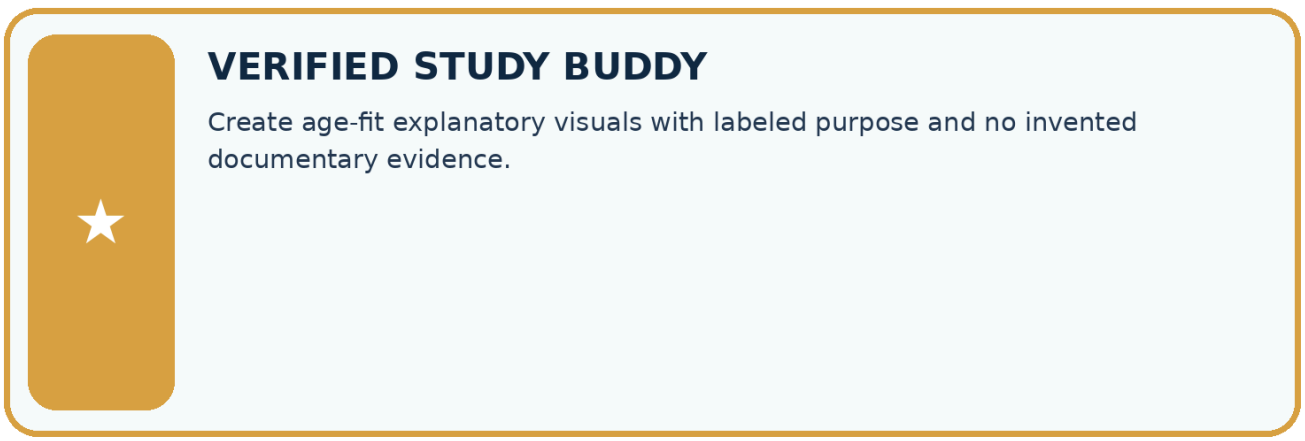
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: show which relationship the evidence supports

> Keep proposed repairs and actually inspected results distinct.

Explain without notes

- Close your notes. Explain why correct object counts do not establish position, full containment or the intended arrow meaning.

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
subject	The main person, object or topic represented in an image or piece of content.	
setting	A chosen configuration value that affects how a system operates.	
framing	The way a question or situation is presented, which can influence interpretation.	
relationship	A connection between quantities, events or concepts; an association need not be causal.	

Terms for the connected explanation

Word	Meaning
Provenance	Recorded origin and transformation history

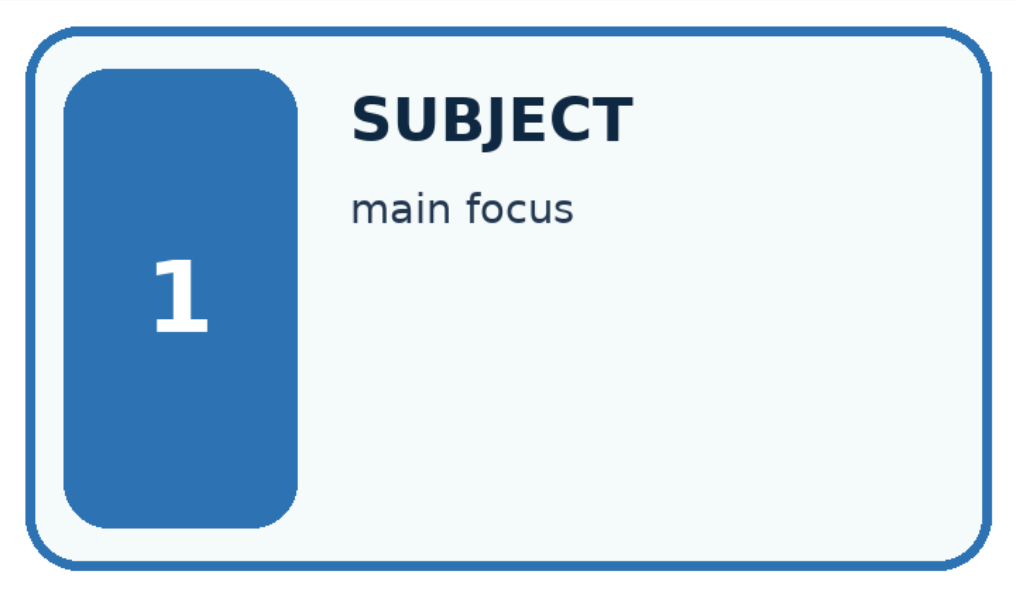
Word	Meaning
Visual criterion	A stated requirement checked against an image
Authenticity claim	A claim about the source or reality of depicted content

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

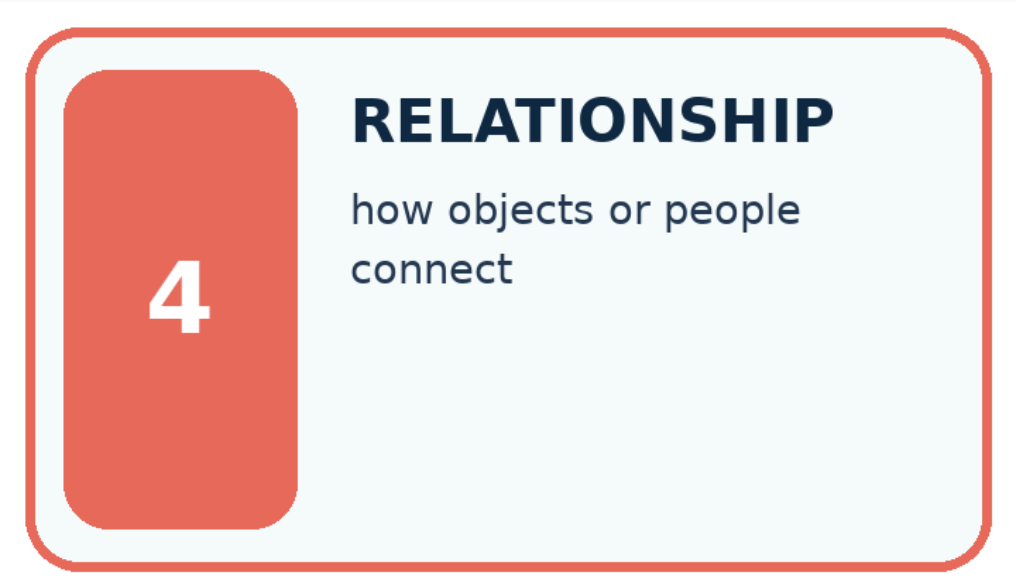
SETTING

environment around the subject

3

FRAMING

viewpoint and crop



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

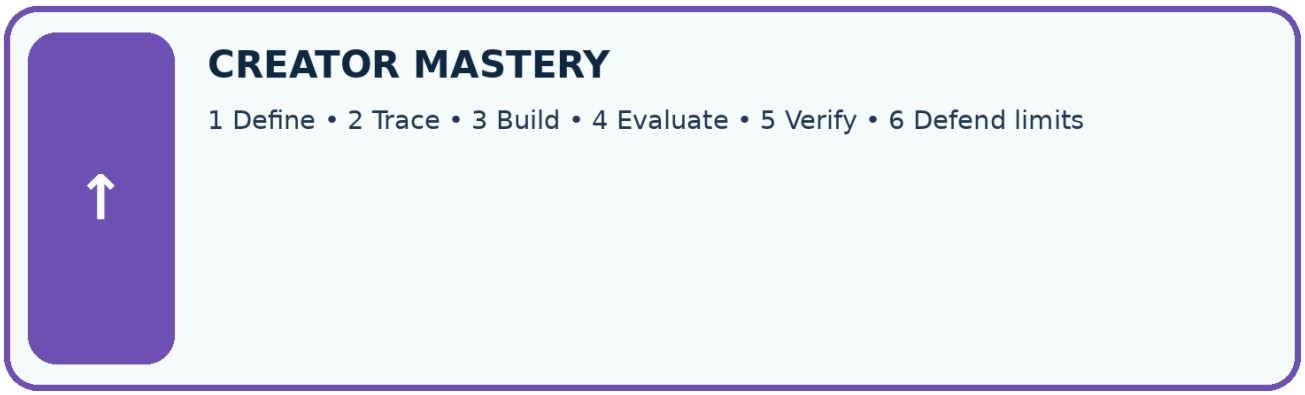
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit a new spatial description independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

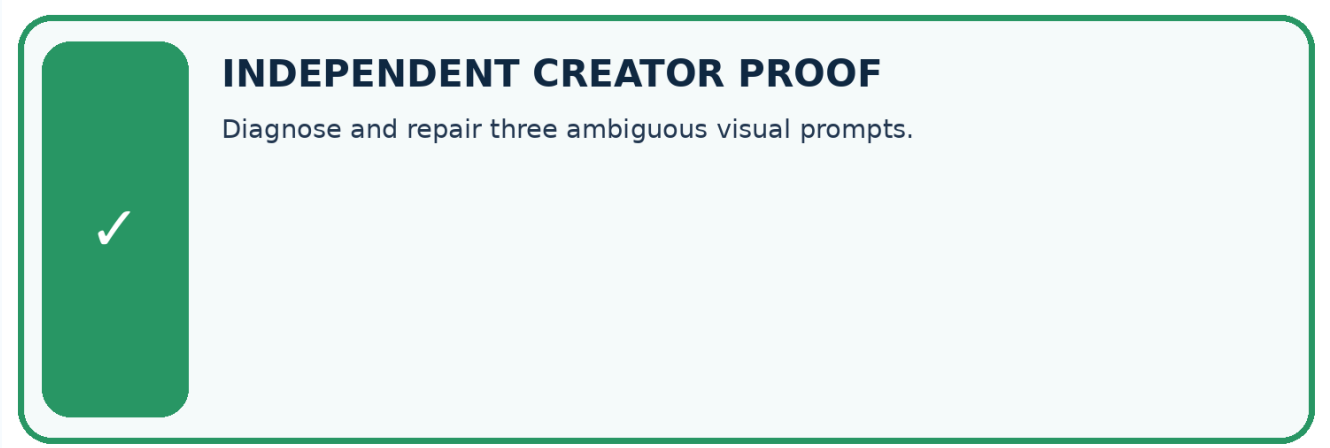
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit a new spatial description independently is the selected independent case.



EXPLANATION CHECK Require count and relation checks with evidence from the supplied description; do not imply inspection of an unseen output.

Write or draw your reasoning and evidence.

▼ **Extra practice: Map the brief to regions**

REFERENCE · SECTION 13

Extra practice: Map the brief to regions

> Try the question before opening its worked explanation.

Read the supplied case and question

Brief: large tree left, small bicycle right, open sky above. **Output** puts bicycle left.

Create a visual checklist.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Check **subject**s, relative size, positions and sky separately. The position violates composition even if **subject**s exist.

Evidence or stage	Value or result
Upper region	Open sky
Left foreground	Large tree
Right foreground	Small bicycle

▼ Earlier wording, alternatives and picture transcripts

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Map the brief to regions

Brief: large tree left, small bicycle right, open sky above. **Output** puts bicycle left.

Create a visual checklist.

Check **subject**s, relative size, positions and sky separately. The position violates composition even if **subject**s exist.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of image generation and provenance	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.
- Which labelled step matches this detail? Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.
- 1. Check position
- 2. Repair brief
- 3. Count objects

- 4. Check direction

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Repair brief Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Changed-case question

- Changed-case practice: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. Position now passes; arrow direction still fails.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Position now passes; arrow direction still fails. Changing object placement does not automatically reverse an arrow's semantic source and destination.

Repair and save your work

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and **provenance**.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

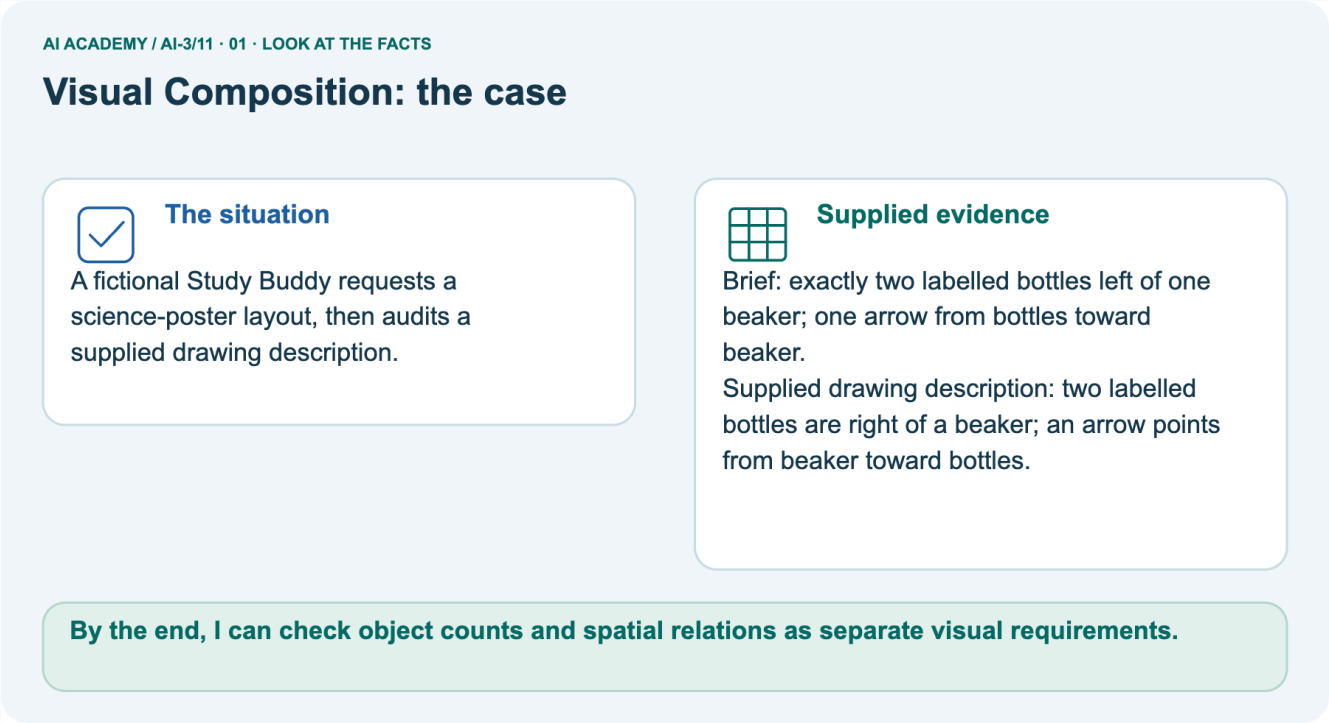
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Visual Composition: the case



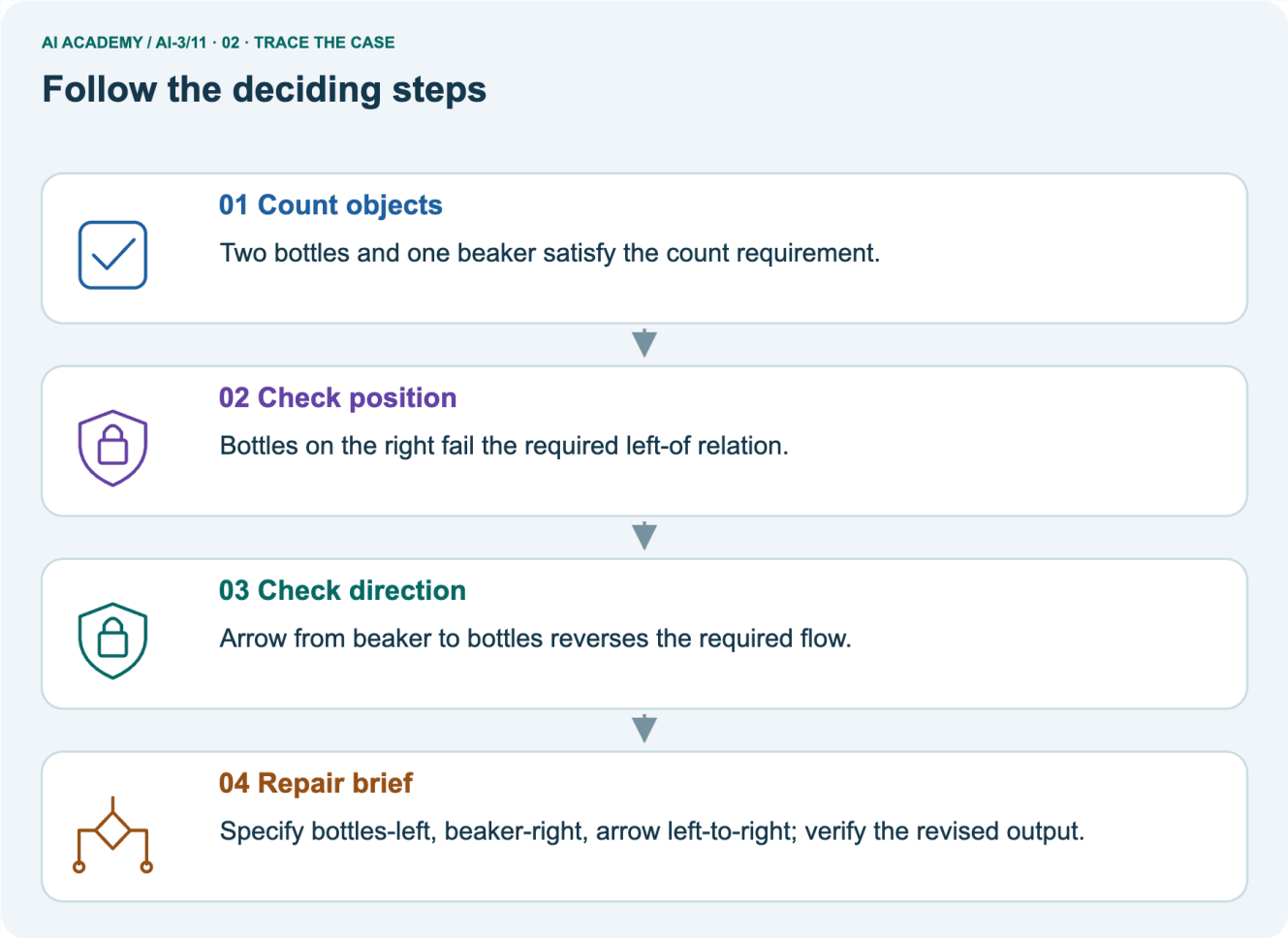
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description. Supplied facts: Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

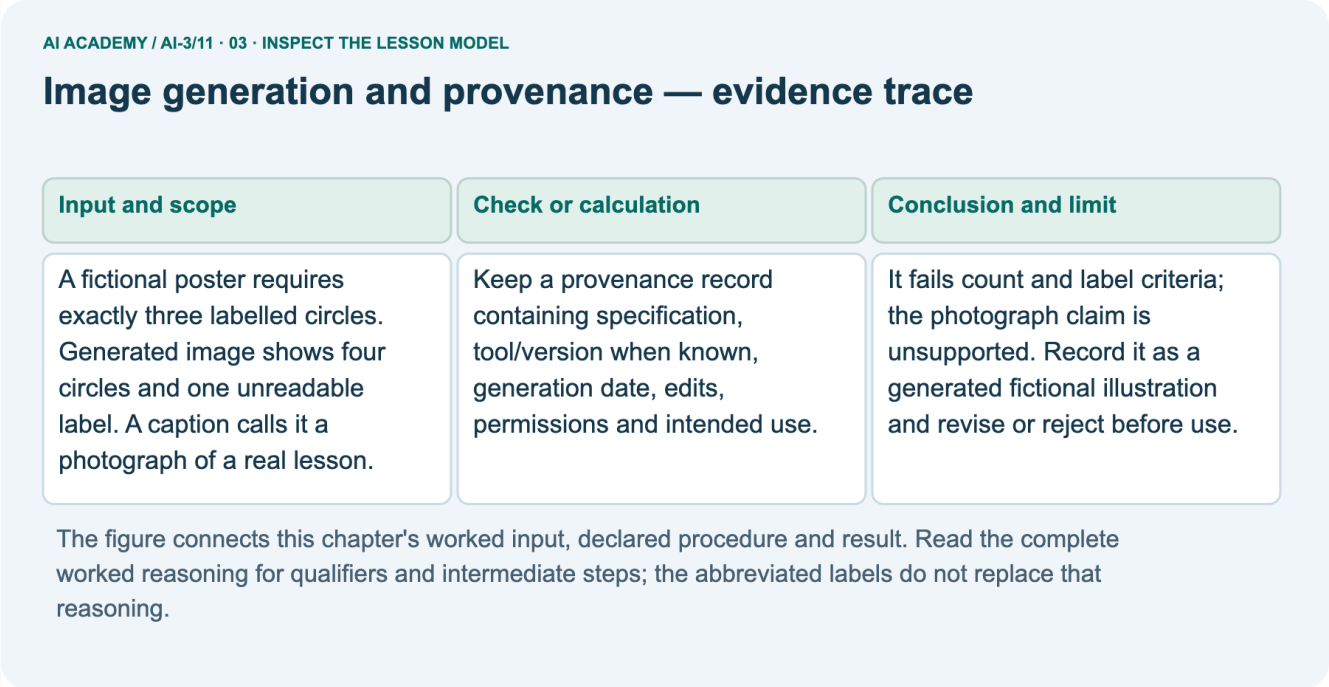


Follow the labelled picture with the matching explanation. Open the vector diagram

Count objects: Two bottles and one beaker satisfy the count requirement. → Check position: Bottles on the right fail the required left-of relation. → Check direction: Arrow from beaker to bottles reverses the required flow. → Repair brief: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Image generation and provenance — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

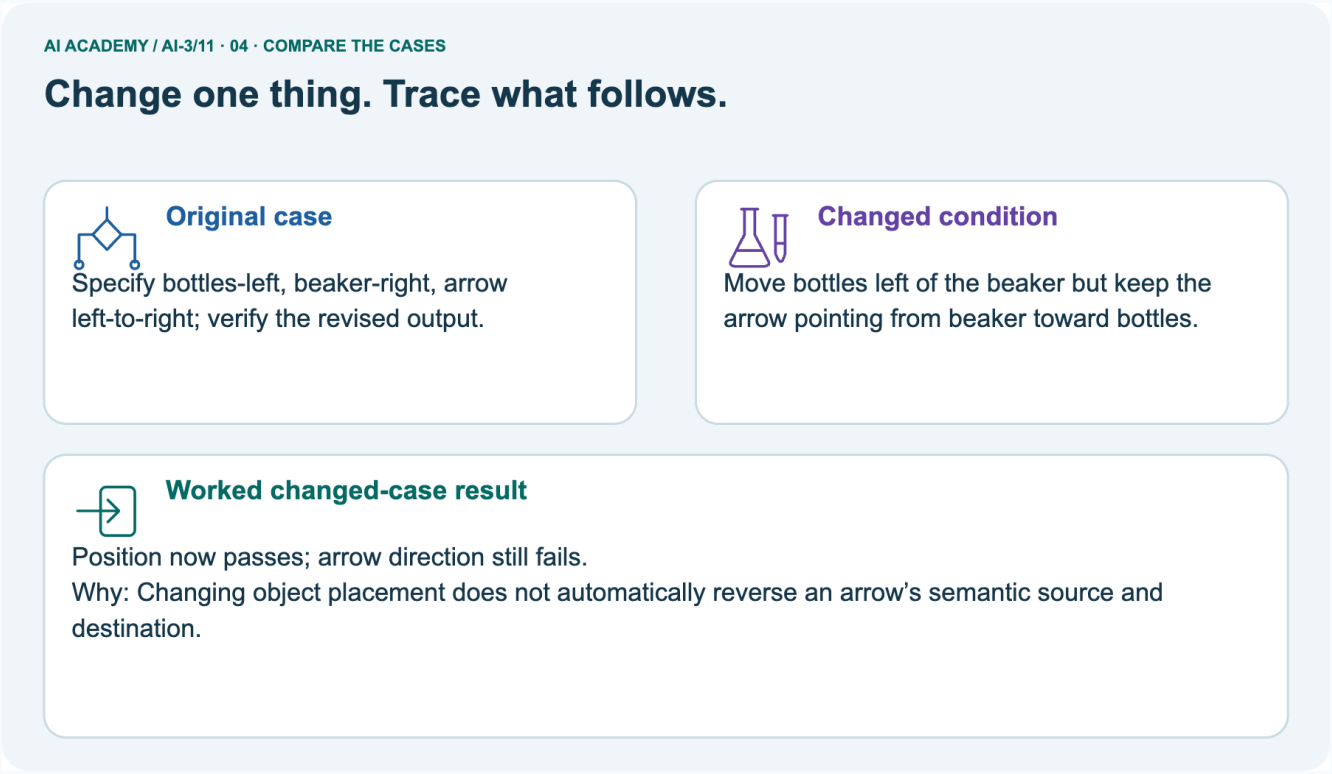
Image generation and **provenance** — **evidence** trace. A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label.

A caption calls it a photograph of a real lesson. | Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. | It fails count and label criteria; the photograph claim is unsupported.

Record it as a generated fictional illustration and revise or reject before use.

Read the labels and values. Explain the **relationship** the model shows.

04. Change one thing. Trace what follows.



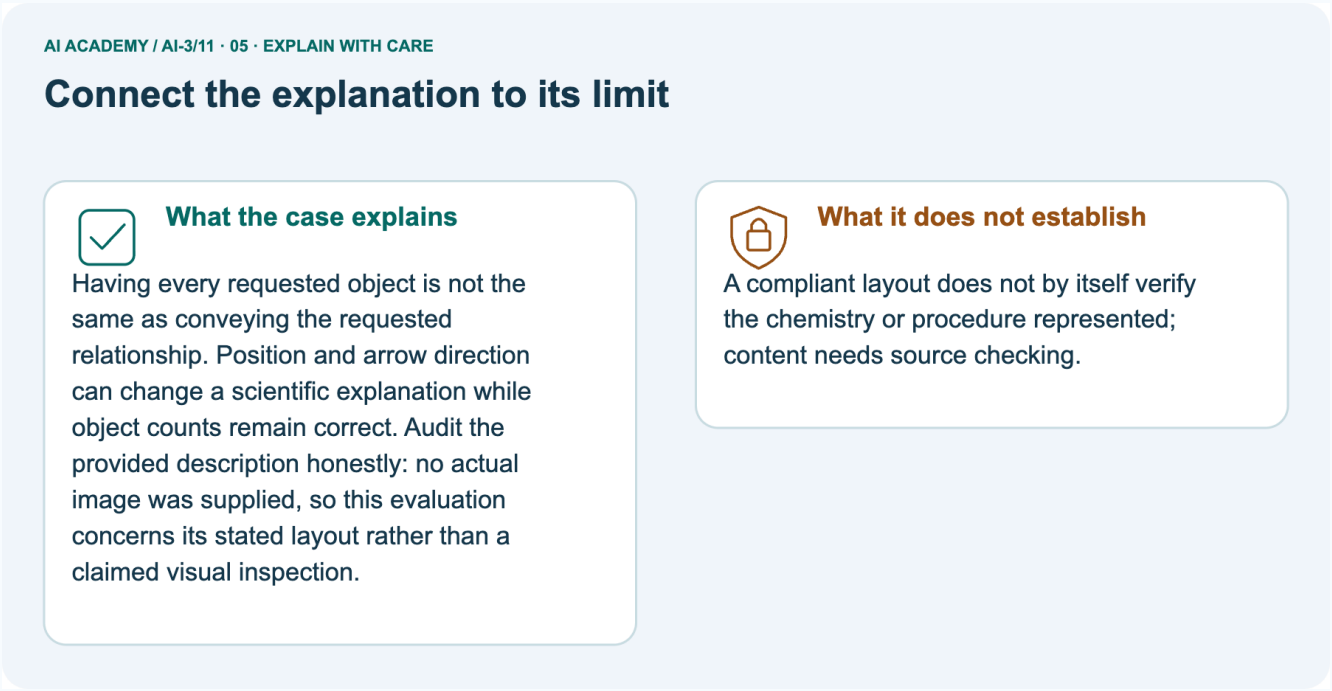
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised **output**. Changed condition: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Expected result: Position now passes; arrow direction still fails. Reason: Changing object placement does not automatically reverse an arrow's semantic source and destination.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

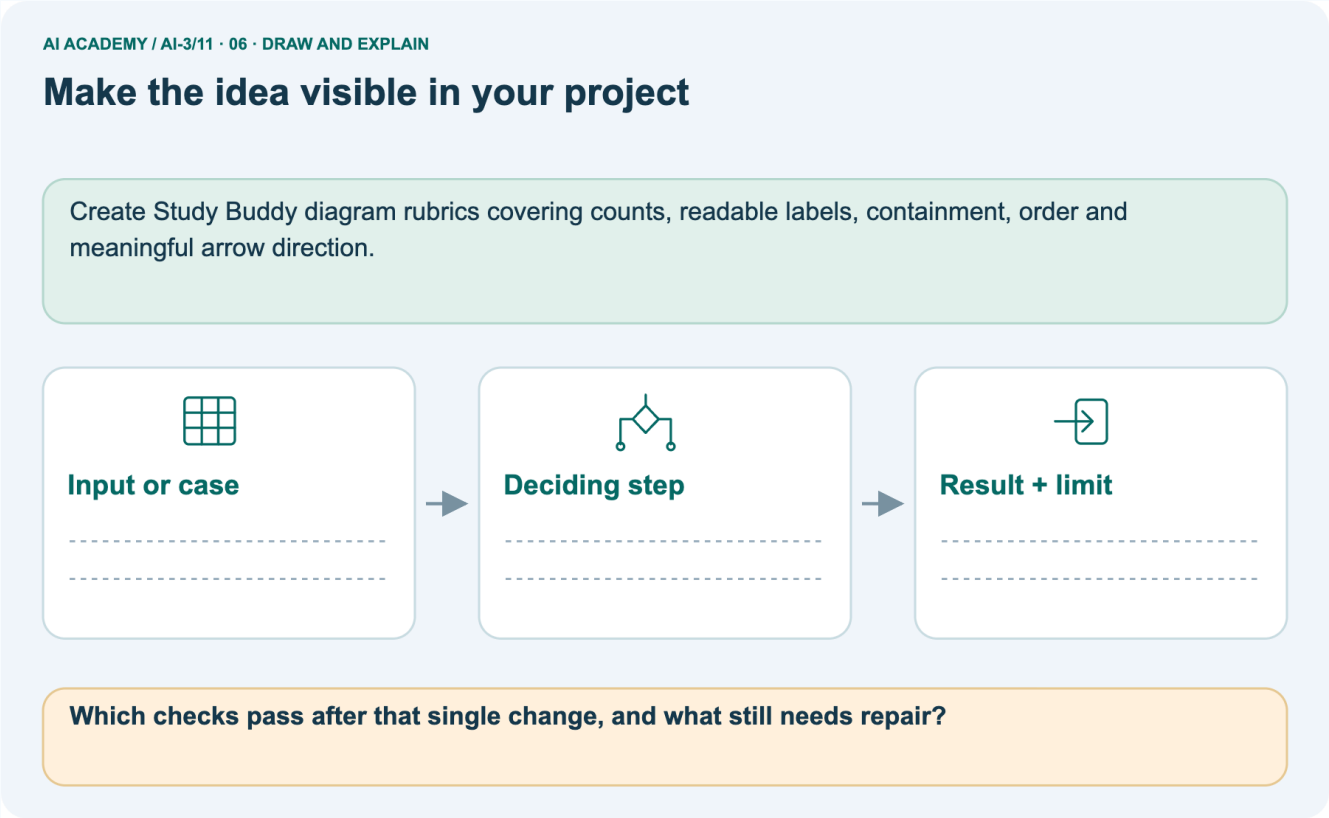
Having every requested object is not the same as conveying the requested **relationship**. Position and arrow direction can change a scientific explanation while object counts remain correct.

Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Limit: A compliant layout does not by itself verify the chemistry or procedure represented; content needs source checking.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction. Draw your **input**, deciding step, result and limit.

Supported question: Which checks pass after that single change, and what still needs repair?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Visual Composition: the case

02. Follow the deciding steps

03. Image generation and provenance — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.